

PASSWORD STRENGTH ANALYZER

JAVA MINI PROJECT

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1. Introduction

The Password Strength Analyzer is a Java-based console application designed to analyze the strength of a password entered by the user. The program checks different characteristics of the password such as length, uppercase letters, lowercase letters, numbers, and special characters.

The application assigns a score based on these conditions and categorizes the password as WEAK, MEDIUM, or STRONG. It also identifies some commonly used passwords and displays a warning to the user.

This project demonstrates the use of Java concepts such as Scanner, strings, conditional statements, regular expressions, and basic score-based logic.

2. Objectives

The main objectives of this project are:

- To analyze the strength of a password.
- To check whether the password has a minimum length of 8 characters.
- To identify uppercase letters.
- To identify lowercase letters.
- To identify numbers.
- To identify special characters.

- To detect a few commonly used passwords.
- To calculate a password strength score.
- To classify the password as Weak, Medium, or Strong.
- To understand practical usage of Java string and conditional operations.

3. Technologies Used

Technology	Purpose
Java	Programming Language
Scanner	Reading user input
String	Storing and processing password
Regular Expressions	Checking password characters
Conditional Statements	Password strength classification
Console	Displaying results

4. System Requirements

Hardware Requirements

- Computer/Laptop
- Minimum 2 GB RAM
- Keyboard and monitor
- Software Requirements
- Java Development Kit (JDK)

- Java compiler
- Any Java IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code
- Operating System: Windows/Linux/macOS

5. Features of the Project

The Password Strength Analyzer provides the following features:

1. Accepts a password from the user.
2. Checks whether the password is commonly used.
3. Checks password length.
4. Checks for uppercase letters.
5. Checks for lowercase letters.
6. Checks for numbers.
7. Checks for special characters.
8. Calculates a score out of 5.
9. Displays the final password strength.

6. Working Principle

The program first displays the title PASSWORD STRENGTH ANALYZER and asks the user to enter a password.

After receiving the password, the program checks whether it matches any of the following common passwords:

- password

- 123456
- qwerty
- admin

The comparison is performed without considering uppercase/lowercase differences.

Next, the program performs five password checks:

1. Length Check

If the password contains at least 8 characters, the program adds one point.

2. Uppercase Check

The program checks whether the password contains at least one uppercase letter from A-Z.

3. Lowercase Check

The program checks whether the password contains at least one lowercase letter from a-z.

4. Number Check

The program checks whether the password contains at least one digit from 0-9.

5. Special Character Check

The program checks whether the password contains at least one character that is not a letter or number.

Each successful condition increases the score by 1.

7. Algorithm

Step 1: Start the program.

Step 2: Display the Password Strength Analyzer heading.

Step 3: Create a Scanner object to read input.

Step 4: Ask the user to enter a password.

Step 5: Convert the password to lowercase and compare it with common passwords.

Step 6: Check whether the password length is at least 8 characters.

Step 7: Check for an uppercase letter.

Step 8: Check for a lowercase letter.

Step 9: Check for a number.

Step 10: Check for a special character.

Step 11: Calculate the score out of 5.

Step 12: Classify the password:

Score 0–2 → WEAK

Score 3–4 → MEDIUM

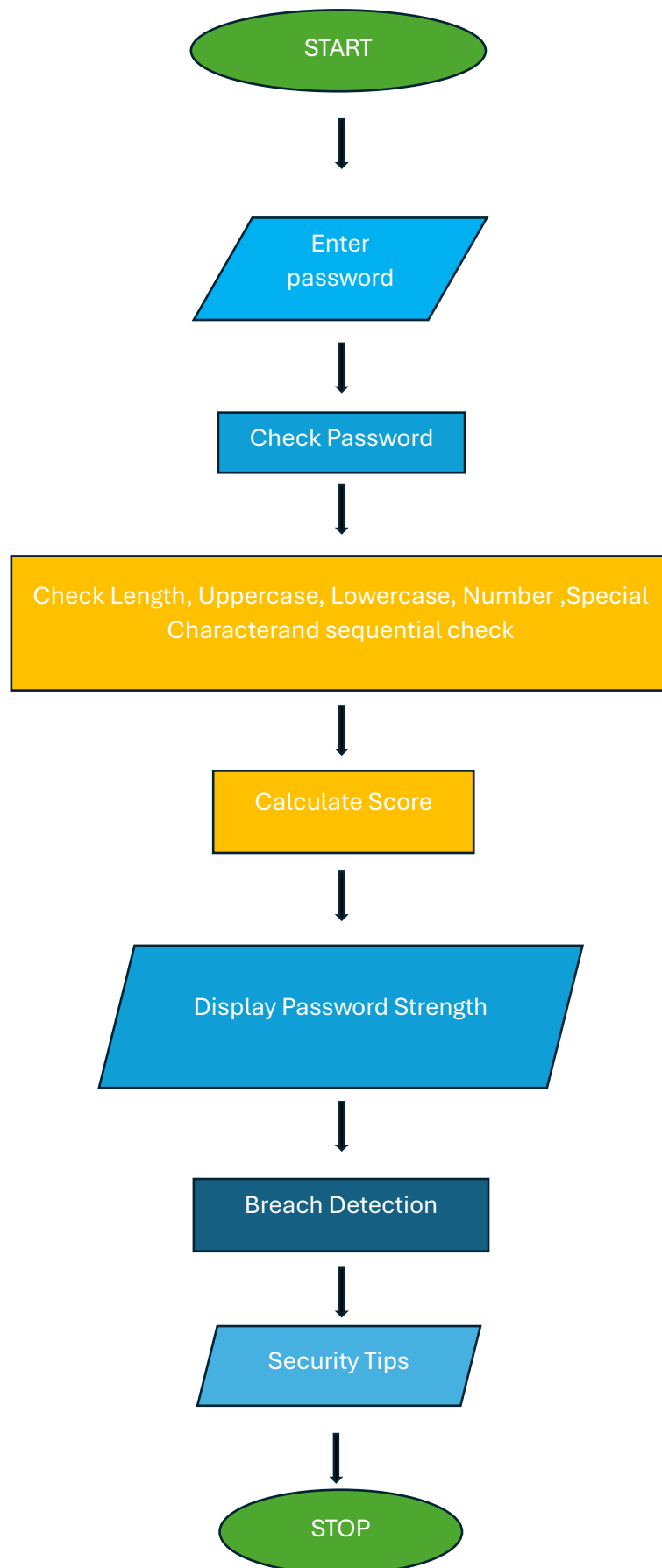
Score 5 → STRONG

Step 13: Display the result.

Step 14: Close the scanner.

Step 15: End the program.

FLOWCHART



8. Program Logic

The scoring system used in the project is:

Condition	Score
Length $\geq$ 8	1
Uppercase letter	1
Lowercase letter	1
Number	1
Special character	1
Maximum Score	5

Strength Classification

Score	Strength
0–2	WEAK
3–4	MEDIUM
5	STRONG

9. Important Java Concepts Used

Scanner

The Scanner class is used to accept the password from the user.

```
Scanner sc = new Scanner(System.in);
```

```
String password = sc.nextLine();
```

String Methods

The `toLowerCase()` method is used to convert the password into lowercase for common-password checking.

```
String lowerPassword =  
password.toLowerCase();
```

Conditional Statements

`if`, `else if`, and `else` statements are used to check conditions and determine password strength.

Regular Expressions

Regular expressions are used to identify different types of characters.

For example:

```
password.matches(".*[A-Z].*")
```

checks for an uppercase letter.

```
password.matches(".*[0-9].*")
```

checks for a number.

10. Sample Output

Example 1 – Strong Password

```
=====
      PASSWORD STRENGTH ANALYZER
=====
Enter your password: Abc@12345

Password Analysis:
-----
? Common Password: Not Detected
? Length: Good
? Uppercase: Good
? Lowercase: Good
? Number: Good
? Special Character: Good
? Sequential Pattern: Detected
-----
Score: 5/5
Password Strength: STRONG
Strength Percentage: 100%

Breach Detection:
Checking known data breaches...
? Breach Detection: FOUND
This password has appeared in known data breaches.
Times seen: 888064
```

Example 2 – Medium Password

```
=====
      PASSWORD STRENGTH ANALYZER
=====
Enter your password: abc12345

Password Analysis:
-----
? Common Password: Not Detected
? Length: Good
? Uppercase: Missing
? Lowercase: Good
? Number: Good
? Special Character: Missing
? Sequential Pattern: Detected
-----
Score: 3/5
Password Strength: MEDIUM
Strength Percentage: 60%

Breach Detection:
Checking known data breaches...
? Breach Detection: FOUND
This password has appeared in known data breaches.
Times seen: 1324706
```

Example 3 – Weak Password

```
=====
      PASSWORD STRENGTH ANALYZER
=====
Enter your password: abc

Password Analysis:
-----
? Common Password: Not Detected
? Length: Too Short
? Uppercase: Missing
? Lowercase: Good
? Number: Missing
? Special Character: Missing
? Sequential Pattern: Detected
-----
Score: 1/5
Password Strength: WEAK
Strength Percentage: 20%

Breach Detection:
Checking known data breaches...
? Breach Detection: FOUND
This password has appeared in known data breaches.
Times seen: 526728
```

11. Advantages

- Simple and easy to understand.
- Easy to execute using Java.
- Checks multiple password characteristics.
- Uses a simple scoring mechanism.
- Helps users understand basic password-strength requirements.
- Demonstrates practical Java programming concepts.

12. Limitations

- The program checks only a small list of common passwords.
- It does not compare the password against a large password database.
- It does not estimate how quickly a password could be guessed.
- It does not check whether a password has appeared in a known data breach.
- The scoring system is a simple educational model and is not a complete measure of real-world password security.
- The password is displayed on the console after entry, which would not be appropriate for a real password application.

13. Future Enhancements

The project can be improved by adding:

A larger common-password database.

Detection of repeated characters and predictable patterns.

Password-breach checking using a privacy-preserving service.

A graphical user interface (GUI).

Password masking while typing.

Password generation functionality.

More detailed password feedback.

A stronger password-strength estimation algorithm.

Secure handling so the entered password is not displayed or stored unnecessarily.

14. Applications

The Password Strength Analyzer can be used as:

A Java programming mini-project.

A demonstration of string processing.

A cybersecurity awareness application.

A learning project for regular expressions.

An educational tool for understanding password requirements.

15. Conclusion

The Password Strength Analyzer is a simple Java console application that evaluates a password using five basic criteria: length, uppercase letters, lowercase letters, numbers, and special characters.

The project successfully demonstrates how Java can be used to process strings, apply conditions, use regular expressions, calculate scores, and display meaningful results. It also provides a basic understanding of password-security principles.

Overall, this project is useful for learning fundamental Java programming concepts while introducing the importance of creating stronger passwords.